

SYDNEY TECHNICAL HIGH SCHOOL



MATHEMATICS

HSC ASSESSMENT TASK 3

JUNE 2008

Time Allowed: 70 minutes

Instructions:

- Write using blue or black pen
- Approved calculators may be used
- Attempt all questions
- All necessary working must be shown. Mark may not be awarded for careless or badly arranged work
- Marks indicated are a guide only and may be varied if necessary
- Start each question on a new side of a page
- A table of standard integrals is supplied

Name:

Q1	Q2	Q3	Q4	Q5	Total

Question 1 (11 marks)**Marks**

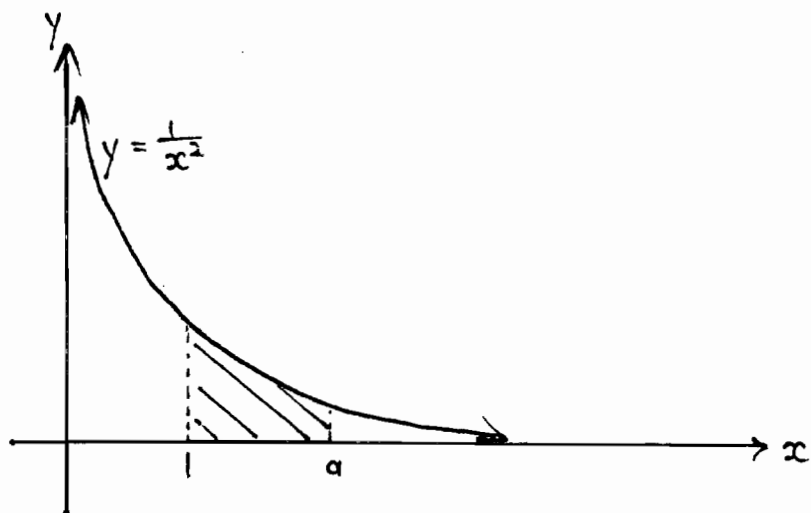
- | | | |
|----|--|---|
| a) | Write 100° in radians in terms of π | 1 |
| b) | Evaluate $\log_{10} 5$ correct to 3 significant figures | 1 |
| c) | Find $\lim_{x \rightarrow 0} \frac{\sin 2x}{x}$ | 2 |
| d) | Solve $\cos x = \frac{\sqrt{3}}{2}$ for $0 \leq x \leq 2\pi$ | 2 |
| e) | Sketch $y = 2\sin(\pi x)$ over the domain $0 \leq x \leq 2$ | 2 |
| f) | If $\log_4 Y = 3.22$ evaluate $\log_4 4Y$ | 2 |
| g) | Find the exact value of $\sin \frac{7\pi}{4}$ | 1 |

Question 2 (11 marks)

- | | | |
|-------|-------------------------------------|---|
| a) | Differentiate with respect to x : | |
| (i) | $y = e^{3x}$ | 1 |
| (ii) | $y = \cos(1 - x^2)$ | 2 |
| (iii) | $y = \log_e \frac{x^2+1}{x}$ | 2 |
| (iv) | $y = e^x \sin x$ | 2 |
| (v) | $y = 10^x$ | 1 |

b)

3



The shaded area above is equal to $\frac{2}{3} \text{ unit}^2$. Find a

Question 3 (11 marks)

a) Find

(i) $\int 2 + \frac{3}{x} dx$ 1

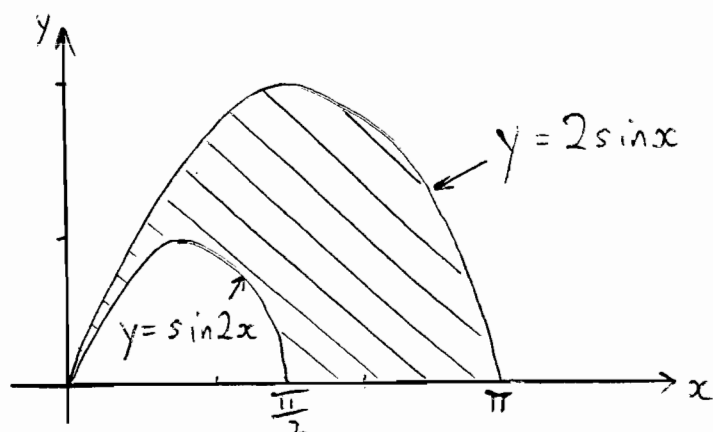
(ii) $\int \sec^2(6x + 1) dx$ 1

(iii) $\int 3e^{2x} dx$ 1

(iv) $\int_{\frac{\pi}{2}}^{\pi} \cos \frac{x}{2} dx$ (exact value) 3

- b) Calculate the area of the shaded region below.

3



- c) By writing $\operatorname{cosec} x$ as $(\sin x)^{-1}$.

Show that $\frac{d}{dx}(\operatorname{cosec} x) = -\operatorname{cosec} x \cot x$

2

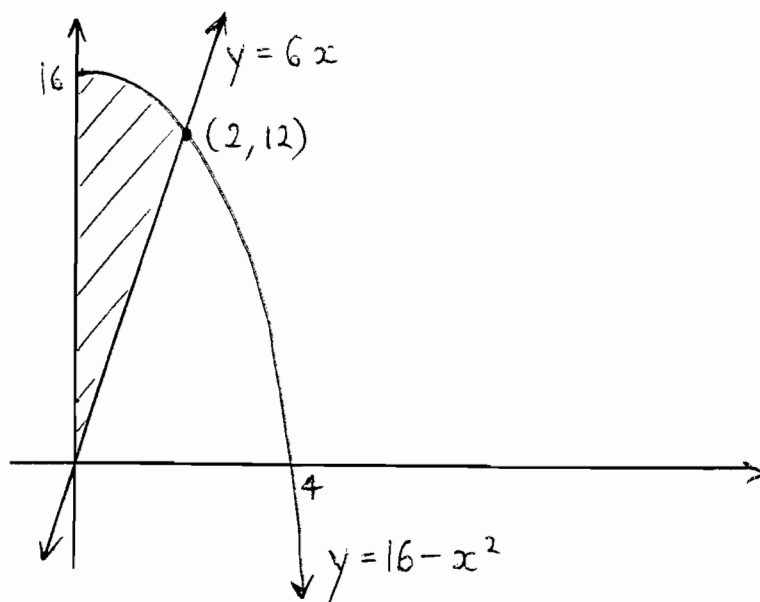
Question 4 (11 marks)

- a) Find $\int \sin\left(\frac{\pi}{4} - x\right) dx$

2

- b)

3



The region above is rotated around the y axis. Find the volume of the solid formed to the nearest whole number.

c) Evaluate $\int_0^{\frac{\pi}{3}} \frac{1+\cos^3 x}{\cos^2 x} dx$ 3

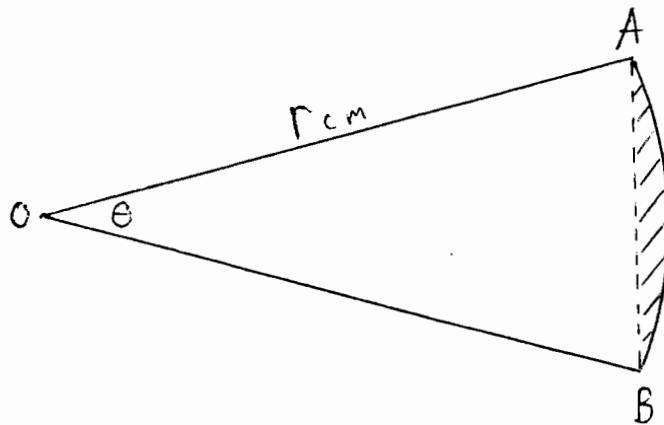
d) (i) Show that $\frac{d}{dx} (x \log_e x) = 1 + \log_e x$ 1

(ii) Hence find $\int \log_e x dx$ 2

Question 5 (11 marks)

Marks

a) The sector OAB below has an area of $2\pi cm^2$. The arc has length $\frac{\pi}{2} cm$.



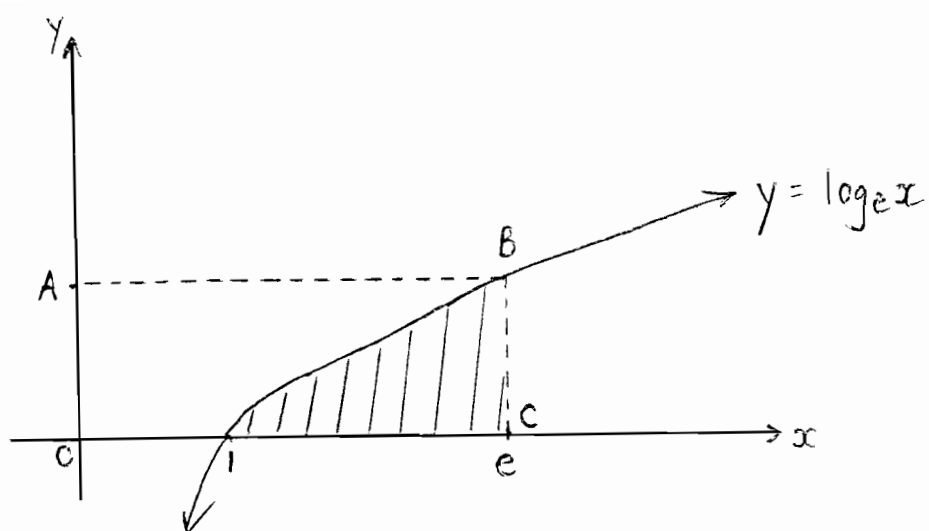
(i) Use this information to form 2 equations. 2

(ii) Hence solve these equations simultaneously to find r and θ 2

(iii) Now find the area of the minor segment shaded above 2

correct to 2 decimal places

b)



- | | | |
|-------|---|---|
| (i) | Using the graph above find the y value at point B | 1 |
| (ii) | Hence find the area of rectangle ABCO. | 1 |
| (iii) | Hence or otherwise find the shaded area. | 3 |

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x$, $x > 0$

Teacher's Name:

Student's Name/N°:

Solutions to 2008 Yr 12 2 Unit Ass. Task 3Question 1

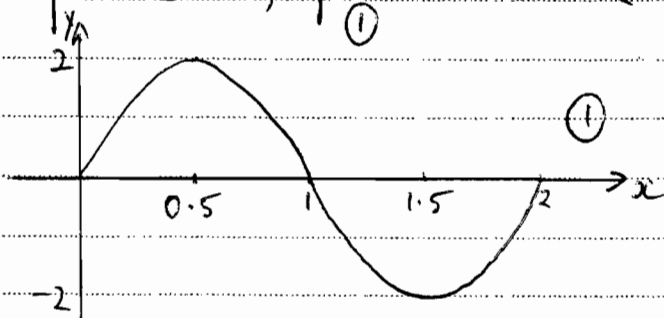
$$a) 100^\circ = 100 \times \frac{\pi}{180} \\ = \frac{5\pi}{9} \text{ radians } \textcircled{1}$$

$$b) \log_{10} 5 = 0.6987 \\ = 0.699 \text{ to 3 s.f.} \textcircled{1}$$

$$c) \lim_{x \rightarrow 0} \frac{\sin 2x}{x} \\ = 2 \lim_{x \rightarrow 0} \frac{\sin 2x}{2x} \\ = 2 \text{ since } \lim_{x \rightarrow 0} \frac{\sin x}{x} = 1. \textcircled{2}$$

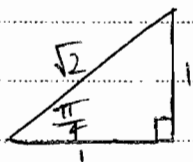
$$d) \cos x = \frac{\sqrt{3}}{2} \\ x = \frac{\pi}{6} \text{ working angle} \\ \begin{array}{c|c} S & A \\ \hline T & C \end{array} \\ x = \frac{\pi}{6}, \frac{11\pi}{6} \textcircled{2}$$

$$e) y = 2 \sin \pi x \\ \text{amplitude} = 2, \text{ period} = \frac{2\pi}{\pi} = 2. \textcircled{1}$$



$$f) \log_4 4Y \\ = \log_4 4 + \log_4 Y \textcircled{1} \\ = 1 + 3.22 \\ = 4.22 \textcircled{1}$$

$$g) \sin \frac{7\pi}{4} \\ = -\sin \frac{\pi}{4} \textcircled{1} \\ = -\frac{1}{\sqrt{2}} \textcircled{1}$$

Question 2

$$i) y = e^{3x} \\ y' = 3e^{3x} \textcircled{1}$$

$$ii) y = \cos(1-x^2) \\ y' = -2x \sin(1-x^2) \textcircled{2}$$

$$iii) y = \log_e \frac{x^2+1}{x} \\ y' = \log_e(x^2+1) - \log_e x \\ y' = \frac{2x}{x^2+1} - \frac{1}{x} \textcircled{2}$$

$$iv) y = e^x \sin x \\ y' = e^x \cos x + \sin x e^x \\ y' = e^x (\sin x + \cos x) \textcircled{2}$$

$$v) y = 10^x \\ y' = 10^x \log_e 10 \textcircled{1}$$

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$$b) \int_1^9 \frac{1}{x^2} dx = \frac{2}{3} \quad (1)$$

$$\left(\frac{-1}{x} \right) \Big|_1^9 = \frac{2}{3} \quad (1)$$

$$\frac{-\frac{1}{9} - \left(-\frac{1}{1}\right)}{\frac{-1}{9} - \frac{-1}{1}} = \frac{\frac{2}{3}}{\frac{-1}{3}} = 3 \quad (1)$$

$$ciii) \int 3e^{2x} dx = \frac{3}{2} e^{2x} + c \quad (1)$$

Question 3

$$a) ci) \int 2 + \frac{3}{x} dx = 2x + 3 \log_e x + c \quad (1)$$

$$cii) \int \sec^2(6x+1) dx = \frac{1}{6} \tan(6x+1) + c \quad (1)$$

$$civ) \int_{\frac{\pi}{2}}^{\pi} \cos \frac{x}{2} dx = \int_{\frac{\pi}{2}}^{\pi} \cos \frac{1}{2} x dx \quad (1)$$

$$\left[2 \sin \frac{1}{2} x \right]_{\frac{\pi}{2}}^{\pi} \quad (1)$$

$$\frac{2 \left[\sin \frac{\pi}{2} - \sin \frac{\pi}{4} \right]}{2 \left(1 - \frac{1}{\sqrt{2}} \right)} \quad (1)$$

$$b) \text{Area} = \int_0^{\pi} 2 \sin x dx - \int_0^{\frac{\pi}{2}} \sin 2x dx \quad (1)$$

$$= \left[-2 \cos x \right]_0^{\pi} - \left[-\frac{1}{2} \cos 2x \right]_0^{\frac{\pi}{2}} \quad (1)$$

$$= (-2 \cos \pi - -2 \cos 0) - \left(-\frac{1}{2} \cos \pi - -\frac{1}{2} \cos 0 \right)$$

$$= (2 + 2) - \left(\frac{1}{2} + \frac{1}{2} \right)$$

$$= 3 \quad (1)$$

$$c) \frac{d}{dx} (\operatorname{cosec} x) = \frac{d}{dx} (\sin x)^{-1} = -(\sin x)^{-2} \times \cos x \quad (1)$$

$$= \frac{-\cos x}{\sin^2 x}$$

$$= \frac{-\cos x}{\sin x} \times \frac{1}{\sin x}$$

Question 4

$$a) \int \sin \left(\frac{\pi}{4} - x \right) dx = + \cos \left(\frac{\pi}{4} - x \right) + c \quad (1)$$

Teacher's Name:

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$$\begin{aligned}
 \text{b) } V &= \pi \int_0^{12} x^2 dy + \pi \int_{12}^{16} x^2 dy \\
 &= \pi \int_0^{12} \left(\frac{y}{6}\right)^2 dy + \pi \int_{12}^{16} 16 - y dy \quad (1) \\
 &= \frac{\pi}{36} \left[\frac{y^3}{3} \right]_0^{12} + \pi \left[16y - \frac{y^2}{2} \right]_{12}^{16} \quad (1) \\
 &= \frac{\pi}{36} \left(\frac{1728}{3} \right) + \pi [256 - 128 - (192 - 72)] \quad (1) \\
 &= 75 \text{ units}^3 \quad (1)
 \end{aligned}$$

$$\begin{aligned}
 \text{c) } \int_0^{\frac{\pi}{3}} \frac{1 + \cos^3 x}{\cos^2 x} dx \\
 \int_0^{\frac{\pi}{3}} \sec^2 x + \cos x dx \quad (1) \\
 [\tan x + \sin x]_0^{\frac{\pi}{3}} \quad (1) \\
 \frac{\tan \frac{\pi}{3} + \sin \frac{\pi}{3}}{\sqrt{3} + \frac{\sqrt{3}}{2}} = \frac{3\sqrt{3}}{2} \quad (1) \\
 \text{or } 2.6 / 2.59
 \end{aligned}$$

$$\begin{aligned}
 \text{d) } \frac{d}{dx} (x \log_e x) &= x \times \frac{1}{x} + \log_e x \\
 &= 1 + \log_e x \quad (1)
 \end{aligned}$$

$$\begin{aligned}
 \text{ii) } \log_e x &= \frac{d}{dx} (x \log_e x) - 1 \\
 &\text{from part ci)} \\
 \int \log_e x dx &= \int \frac{d}{dx} (x \log_e x) dx - \int dx \quad (1) \\
 &= x \log_e x - x + c \quad (1)
 \end{aligned}$$

Question 5

$$\begin{aligned}
 \text{ci) } A &= \frac{1}{2} r^2 \theta \quad l = r\theta \\
 2\pi &= \frac{1}{2} r^2 \theta \quad (1) \quad \frac{\pi}{2} = r\theta \quad (1)
 \end{aligned}$$

$$\begin{aligned}
 \text{cii) } \theta &= \frac{4\pi}{r^2} \text{ sub. into} \\
 \frac{\pi}{2} &= r\theta \\
 \frac{\pi}{2} &= r \times \frac{4\pi}{r^2} \\
 r &= 8 \text{ cm} \therefore \theta = \frac{\pi}{16} \quad (1) \quad (1)
 \end{aligned}$$

$$\begin{aligned}
 \text{ciii) } A &= \frac{1}{2} r^2 (\theta - \sin \theta) \quad (1) \\
 &= \frac{1}{2} \times 8^2 \left(\frac{\pi}{16} - \sin \frac{\pi}{16} \right) \\
 &= 0.04 \text{ cm}^2 \quad (1)
 \end{aligned}$$

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b) A + B, $x = e$

$$\therefore y = \log_e e$$

$$= 1 \quad \textcircled{1}$$

ii) Area = $1 \times e$

$$= e \quad \textcircled{1}$$

iii) Area = Rectangle - $\int x dy$

$$= e - \int_0^1 e^y dy \quad \textcircled{1}$$

$$= e - [e^y]_0^1 \quad \textcircled{1}$$

$$= e - [e - e^0]$$

$$= e - e + 1$$

$$= \underline{1 \text{ unit}^2} \quad \textcircled{1}$$